

Platform Builder

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Economic conditions notwithstanding, many readers have written in over the past two months with queries relating to upgrading their ageing PCs. When building a PC, my advice is to look at exactly what you want and buy according to your needs and applications. Blindly opting for a quad-core CPU with 4 GB of RAM doesn't work for everybody.

There are no less than 40 processor models and well over 150 motherboards available today – a wide variety. Once your decision to upgrade (or buy a new machine) is made, ask yourself for what exactly will you use this configuration? This is the most important part of making a good decision. Are you a casual multimedia user or a gamer? Do you do a lot of encoding?

Most casual multimedia and basic home users will

want to look at a budget dual-core processor. Keep aside a budget of around Rs. 4,000 for the processor. Both Intel and AMD have good offerings at similar price points. A couple of AMD entry-level tri-core Phenom CPUs sneak into this price range.

I recommend AMD's Athlon X2 5200+ (2.6 GHz) for those looking for a decent multimedia PC. It's priced at Rs. 2,950 and represents the rock bottom CPU from AMD; for something faster look at the X2 7750 (2.7 GHz) priced at Rs. 3,200. Gigabyte's MA78GM-US2H built around the 780G chipset is a good board for these CPUs. Priced at Rs. 4,750, this board is feature rich and powerful. Alternatively, ASUS has an offering based on the same chipset called the M3A78-EM but this board does not have an HDMI port. Invest in at least 2 GB of DDR2 RAM or 4 GB for Vista. DDR2 is dirt cheap and 4 GB from Kingston or Corsair will cost

as little as Rs. 2,500.

If you're looking for a gaming PC, remember that your processor is not as important as your graphics card and you should spend most of your money on a high-end card, as this is what really enables gaming at higher resolutions and at higher graphics settings. Most users believe a very fast CPU is important, but the CPU should be fast enough not to bottleneck your graphics card. Games are largely single-threaded with the exception of a few titles and while this will change in the future, a fast dual core is much better than a slower clocked quad core. Hardcore gamers should look at Intel's Core 2 Duo E8500 (3.16 GHz) or E8400 (3.0 GHz). These CPUs are priced at Rs. 9,500 and Rs. 8,400, respectively.

The choice of motherboard is more dependent on the choice of video solution. Those looking at multi-GPU setups will be forced to pick from a limited number of chipsets – NVIDIA's nForce 780i and 790i for SLI and Intel's X48 and X38 chipset-based boards for CrossFire. These boards can also get expensive. For an SLI setup, we recommend Zotac's GeForce 780i Supreme for Rs. 9,500. Unfortunately, X38 and X48 boards are costlier, in the price range of Rs. 14,000 or so. Another solution exists, but it's costly albeit a bit more flexible. Intel's Core i7 processors are based on its X58 platform and this solution has support for both SLI and CrossFire – useful for a long term rig where one might want to change GPU solutions. An i7-based system gets costly as the cheapest CPU (Core i7

920) is priced at Rs. 14,500. You must also invest in costly DDR3 memory as the X58 doesn't support DDR2. For a Core 2 Duo platform based on a single GPU solution, we recommend a motherboard based on Intel's P45 chipset. ASUS' P5Q-C is available for Rs. 10,200 and is a superb, feature-rich solution. Even cheaper is Gigabyte's EP45C-DS3R and is as good in terms of quality and features. You can pick it up for Rs. 7,900. Once again we recommend 2 x 2 GB of RAM.

Some power users will want a quad core processor for bragging rights. Others will demand more grunt for rendering and encoding, both of which are multi-threaded applications. If you must have a quad core, look at Intel's Q8200 (2.33 GHz, Rs. 8,400), it's a reasonably fast CPU and can also be used for gaming. The Intel Q9550 (2.83 GHz) is also available for power users and this CPU has a massive 12 MB L2 cache. It's costly though, at Rs. 14,400, but will satisfy almost any user. For a motherboard, the two P45 options from Gigabyte and ASUS remain unchanged. Once again, the fastest quad core is the Core i7, but the cost of a motherboard and DDR3 memory will be prohibitive for most. However, if you must go this way, opt for a Core i7 920 (2.66 GHz), Intel DX58-SO motherboard (Rs. 14,500) and at least 4 GB of DDR3 1,333 MHz memory from Kingston/Corsair/OCZ. Expect to pay at least Rs. 6,000 for a 3 GB, tri-channel kit, while faster kits with speeds of 1,600 MHz will cost close to Rs. 10,000. There are 6 GB kits also available, but we don't recommend these unless you are on a 64-bit operating system that can utilise more than 4 GB of RAM and you really need it.

DDR2 RAM is cheap these days and nearly everyone can afford 4 GB. DDR3 prices have also reduced compared to six months back

